



HomeLab Diagrams

Platform Overview

Host Hardware

CPU: Intel Core Ultra 9 285K

Motherboard: Asus ROG Strix Z890I mITX

Memory: G.Skill 96GB DDR5-6400 CL32

Drives: Samsung 990 Pro 2TB / 9100 Pro 4TB

NICs: 3 x 2.5GBE + WIFI 7

GPU: Intel Arc A770



Proxmox Virtual Environment r9.1.4

pve.home.com

192.168.1.178 (mgmt)

192.168.100.10 (WIFI/Lab_LAN1)

192.168.200.178 (Lab_LAN2)

192.168.2.178 (Ext_LAN)

192.168.3.178 (LAG/ISO_LAN2)



PROXMOX



dockerm2
Debian 13
192.168.1.166



dc01.home.com
Windows 2022
Server
192.168.1.152



ks3-control
Red Hat EL 10
192.168.200.21



pfSense-a
FreeBSD 12
192.168.100.1



web.home.com
Ubuntu 25.04
192.168.1.108



trfk.home.com
Debian 12
192.168.1.147

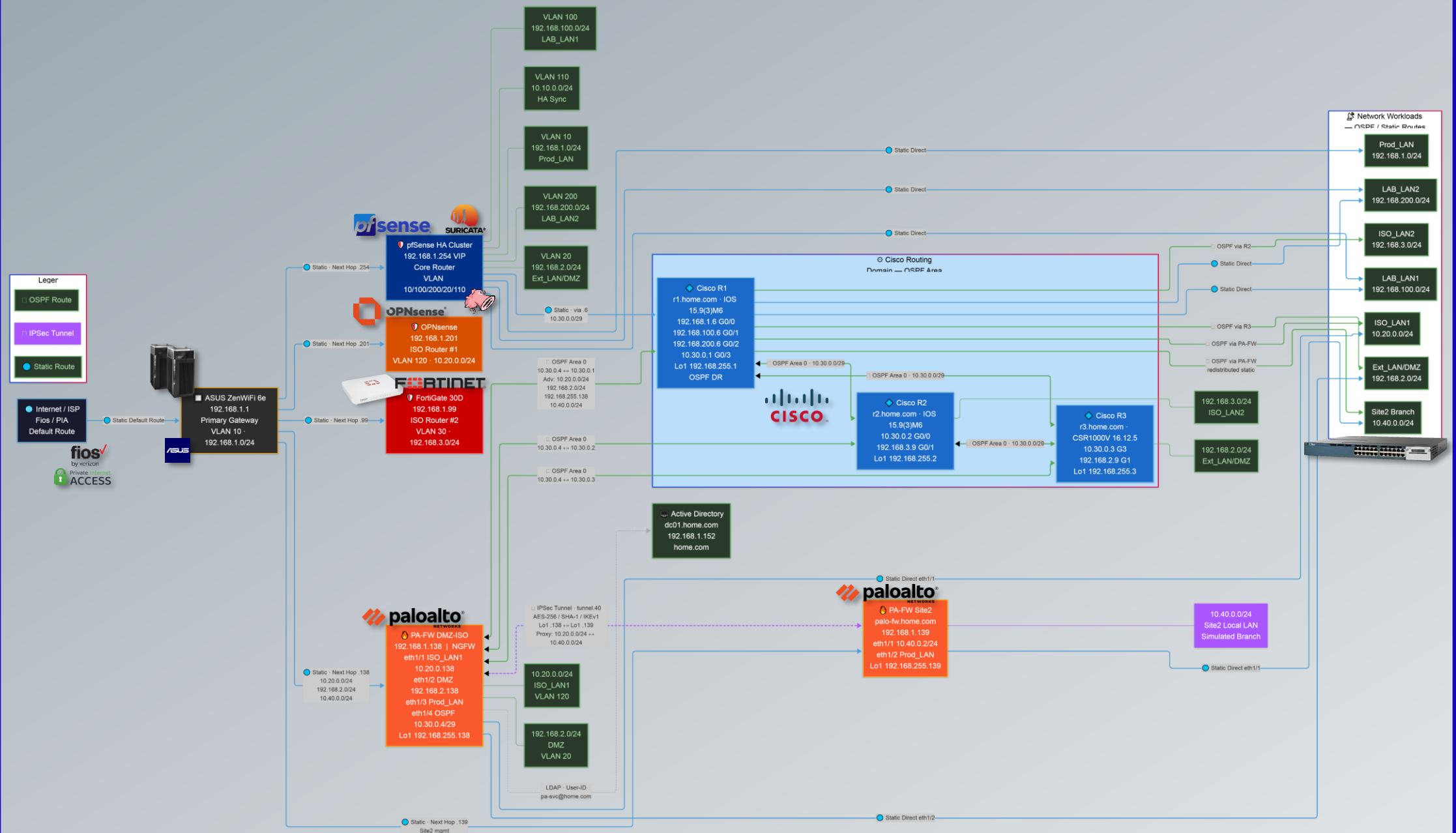


esxi.home.com
VMware ESXi
8.0.3
192.168.1.15



r1.home.com
Cisco iOS 15.9
192.168.100.6

Network Routing Overview



Cisco Overview



CISCO

R2.home.com
iOS 15.9(3)M6
OSPF Area 0
192.168.3.0 0.0.0.255
192.168.2.0 0.0.0.255

192.168.3.0 /24

G0/1 .9



192.168.2.0 /24



Cisco-host2
192.168.3.5
Ubuntu 25.10
LXC



G0/2 .9

Proxmox
vSwitches

192.168.1.0 /24

.2

10.30.0.0 /30

G0/0

.1

G0/3

192.168.200.0 /24



R1.home.com
iOS 15.9(3)M6
OSPF Area 0
192.168.1.0 0.0.0.255
192.168.100.0 0.0.0.255
192.168.100.0 0.0.0.255

192.168.100.0 /24

.6

G0/1



Proxmox
vSwitches

Cisco-host1
192.168.100.4
Ubuntu 25.10
LXC



.6

G0/2

Docker Overview



dockervm2
Debian 13
192.168.1.166

Containers

Portainer-agent
Authentik
OpenVAS
n8n
Watchtower
Elastic-Agent
Metricbeat
PostgreSQL (2)
Redis



ubuntuv1
Ubuntu 25.04
192.168.1.126

Containers

Pihole-bk
Prometheus
Alert Manager
Blackbox
Pulse
Pve-exporter
(Prometheus)
Portainer
Checkmk
WUD
Cloudflared
Discord Proxy
Nebula-sync



trfk.home.com
Debian 12
192.168.1.147

Containers

Traefik
Portainer-agent
Authentik-Outpost
Watchtower
Cloudflared
Cloudflare-DDNS



pihole.home.com
Ubuntu 22.04
192.168.1.250

Containers

Pihole
Portainer-agent
Watchtower
Pihole Exporter
(Prometheus)



Safeline-waf
Debian 13
192.168.1.89

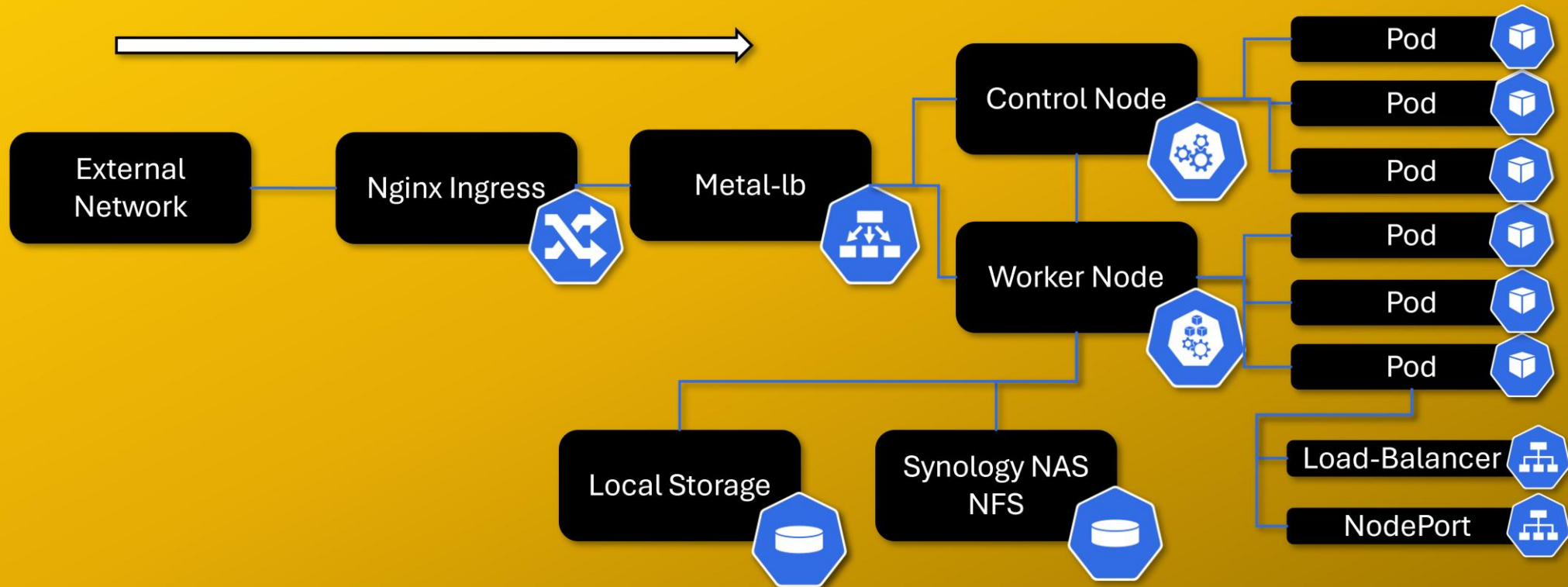
Containers

Safeline Stack
Portainer-agent
SMTP Relay
Netaletx
Watchtower



docker

Lab K3s Topology



VMware Overview



esxi-host1
192.168.100.15
VMware ESXi 8.0.3
(Build 24677879-
standard)

OfficePC2023
192.168.1.31
VMware
Workstation Pro
25H2 (Build
24995812)



Debian 13 Live
192.168.100.220



Windows 2019
Server – Hyper-V
192.168.100.213



pbs.home.com
Proxmox Backup
Server
192.168.1.143



dcm.home.com
Proxmox DC
Manager
192.168.1.47



pmg.home.com
Proxmox Mail
Gateway
192.168.1.57



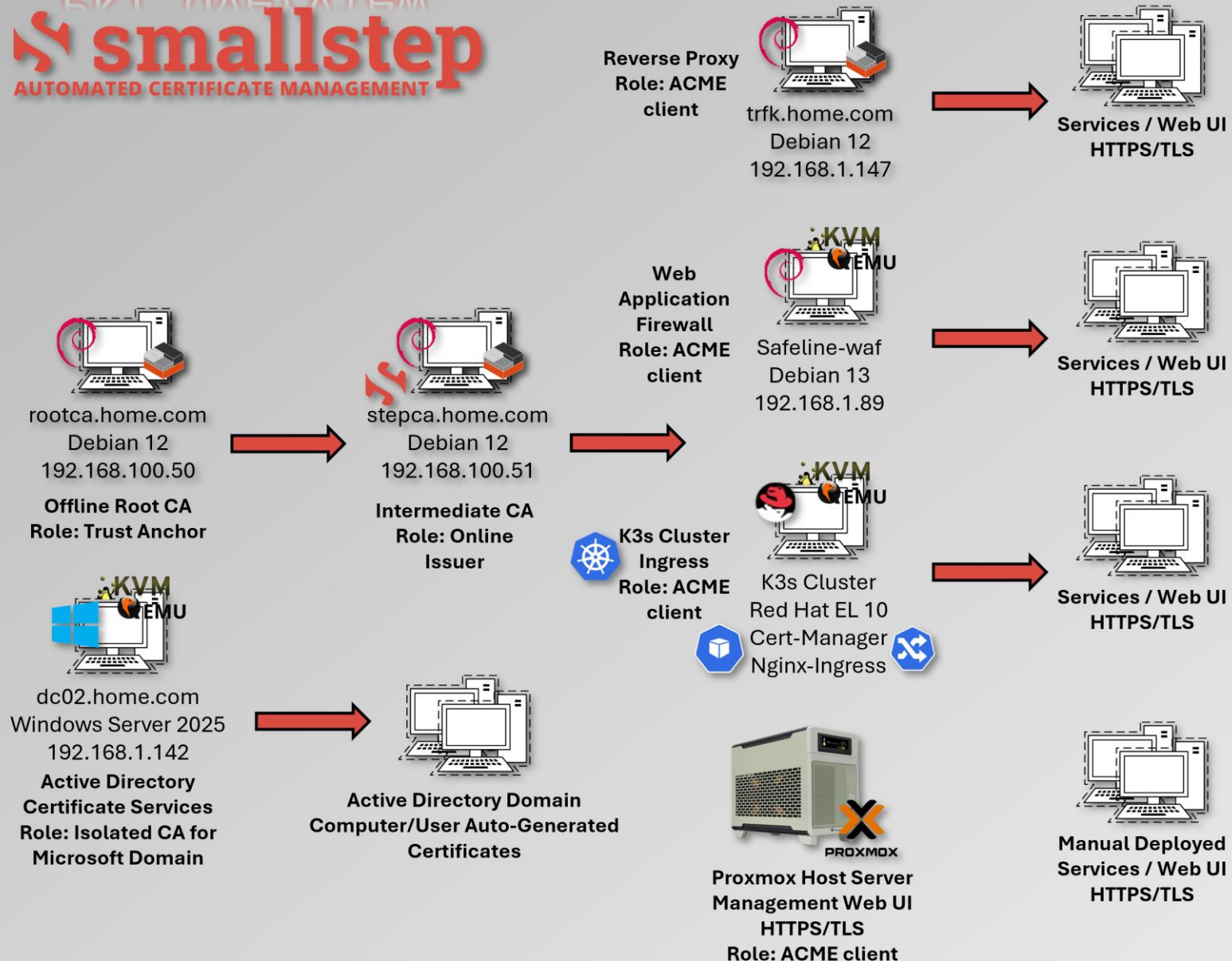
REMnux Linux
192.168.1.56

vmware®

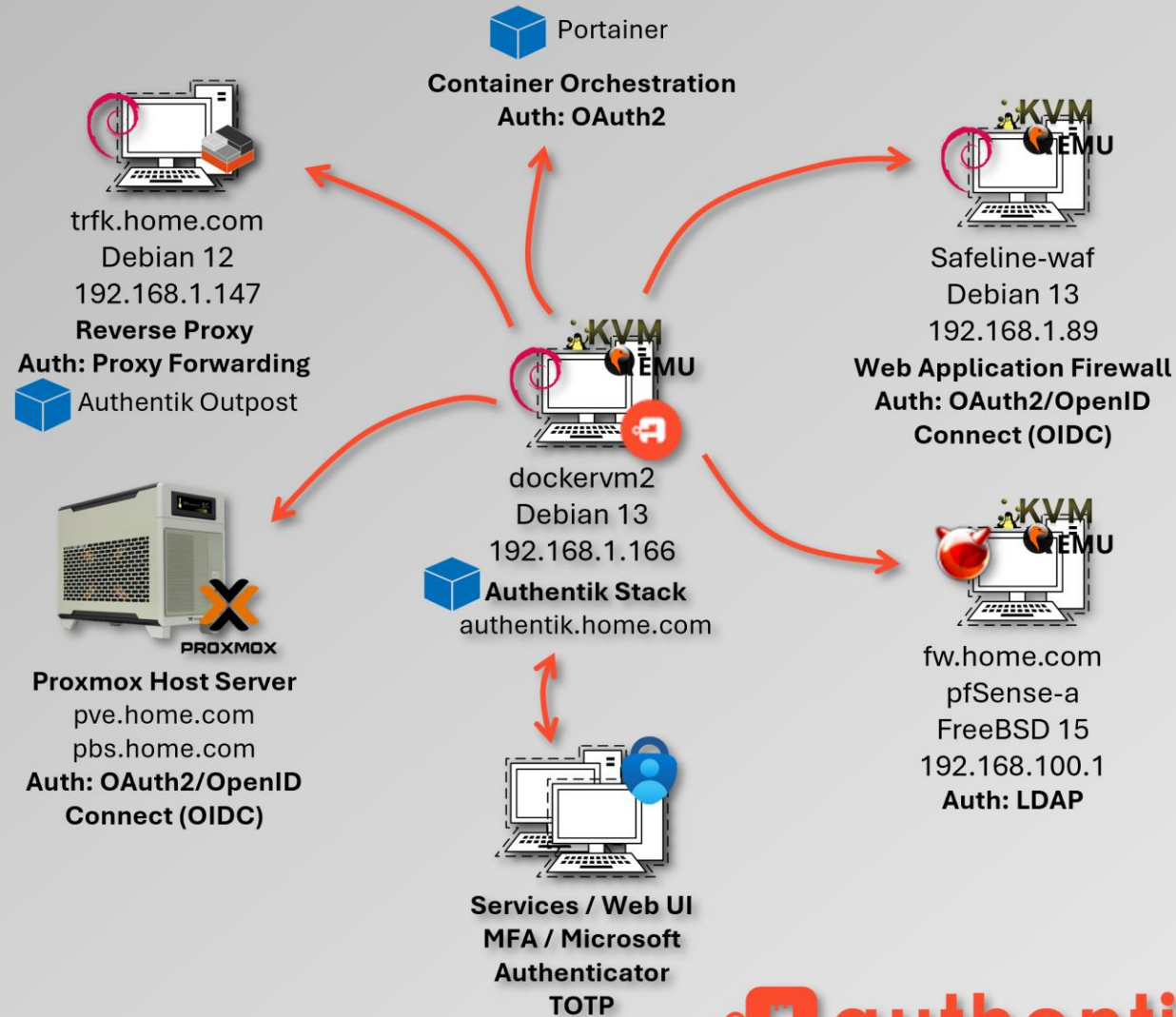
PKI Overview

smallstep

AUTOMATED CERTIFICATE MANAGEMENT



Identity and Access Overview



 **authentik**

Microsoft Active Directory Overview



Windows 2022 Server
192.168.1.152
dc01.home.com
Primary Domain Controller

Deployed Services

Active Directory Domain Services
DNS Server
IIS Webserver
File and Storage Services
Windows Server Update Services



Windows 2025 Server
192.168.1.142, 192.168.1.90
dc02.home.com
Redundant Domain Controller

Deployed Services

Active Directory Domain Services
Active Directory Certificate Services
DNS Server
Hyper-V Manager
IIS Webserver
File and Storage Services
Network Policy and Access Services



Windows 11 Pro
192.168.1.111
win11pro.home.com



Windows 11 Pro WS
192.168.1.184
win11pro2.home.com

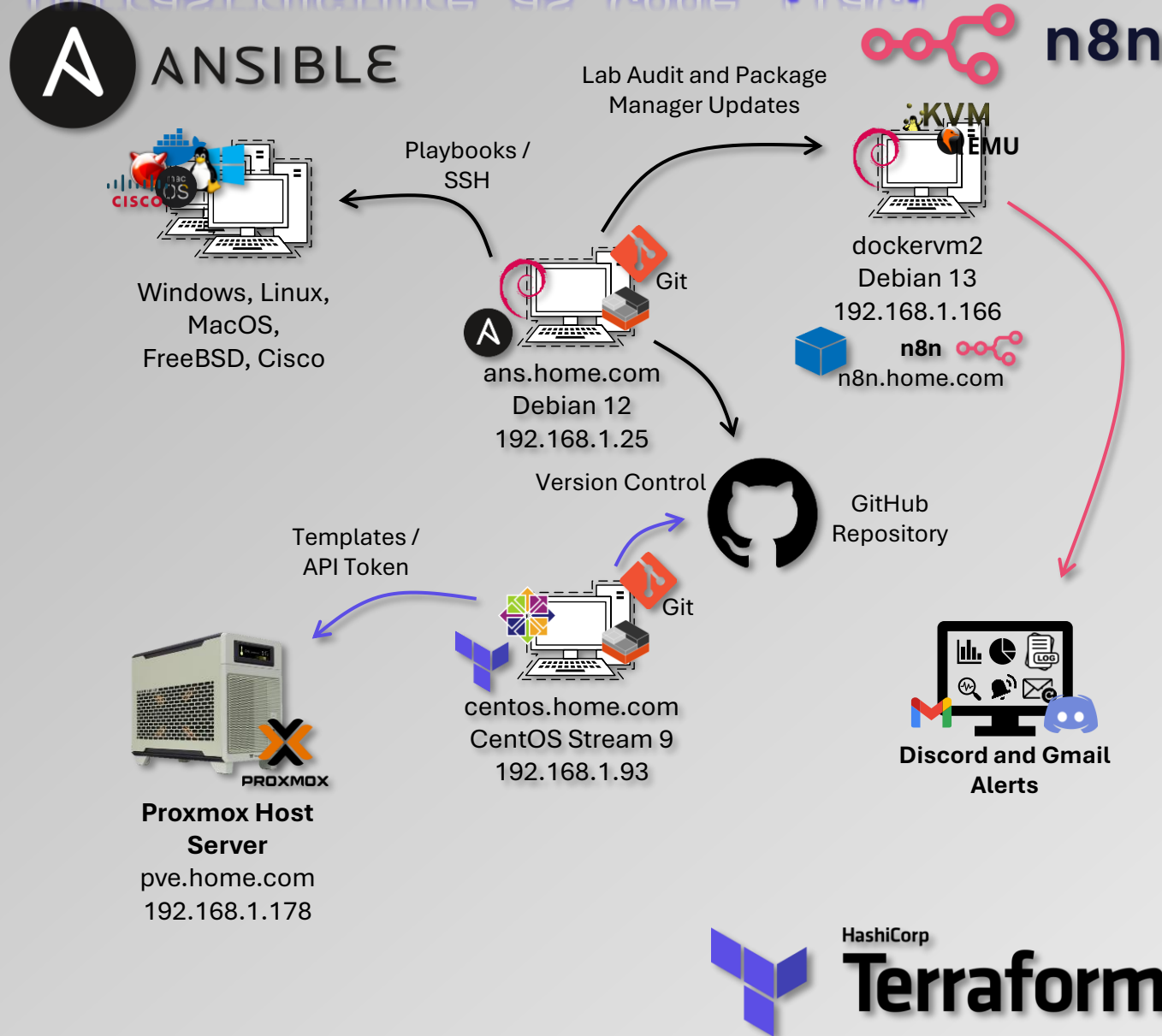


Windows 11
Enterprise
192.168.1.183
win11ent.home.com

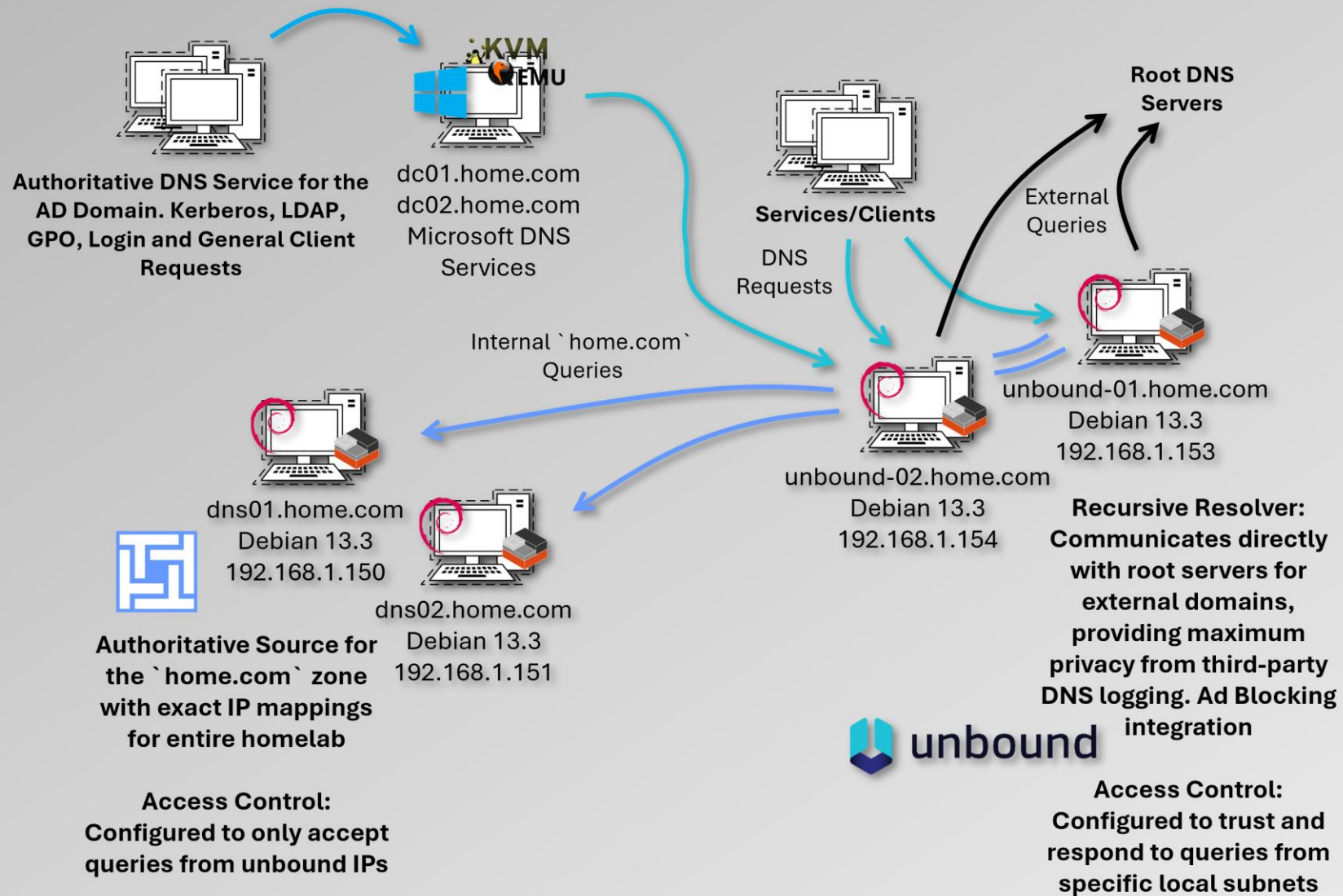


Active Directory

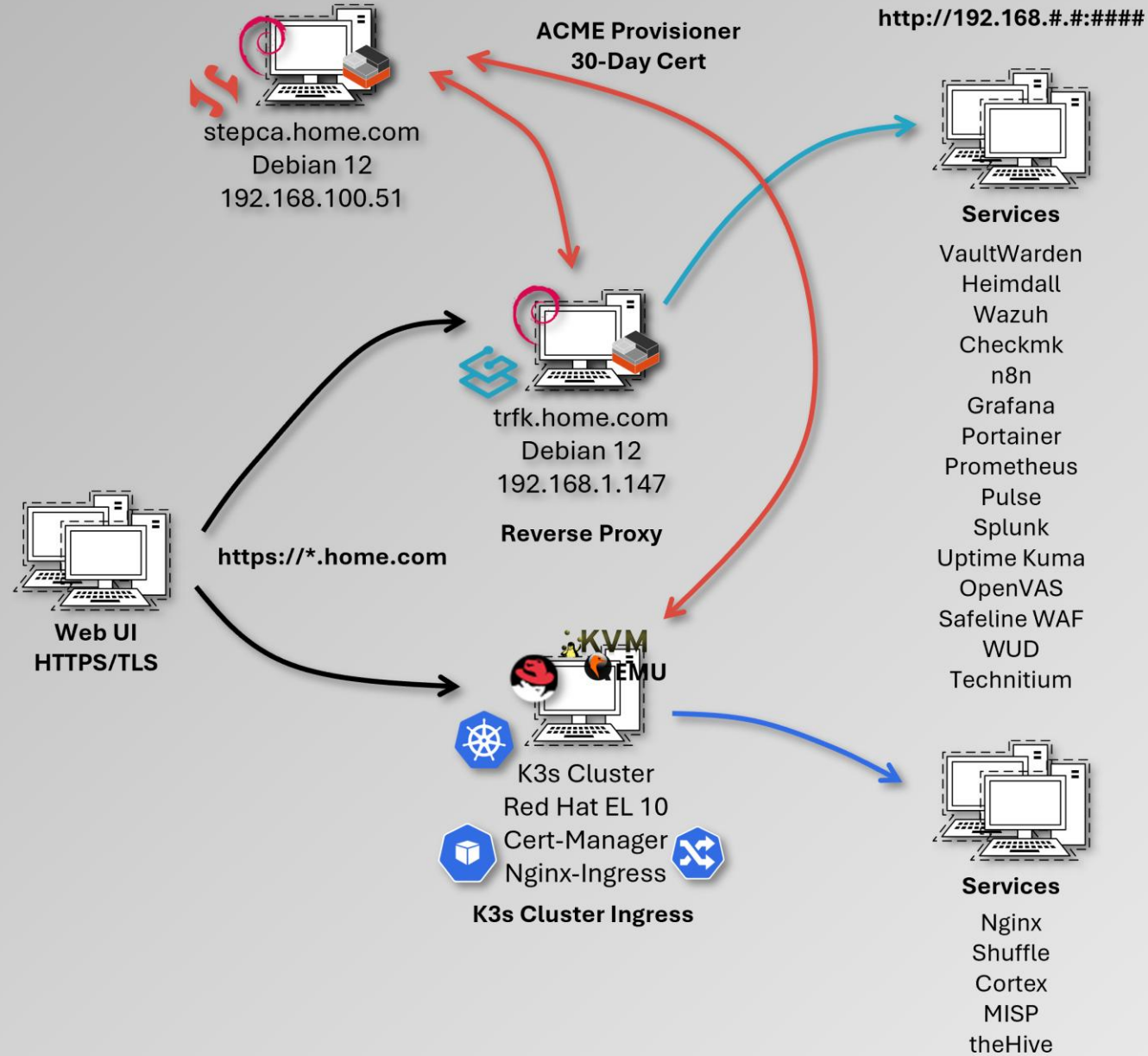
Infrastructure as Code (IaC)



DNS Overview



Reverse Proxy and Ingress Overview



Logging Overview



Elastic Agents
Connected to
Fleet Server



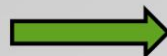
elastic.home.com
Debian 13
192.168.100.8
ELK Stack SIEM
Logging and Alerting



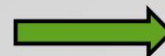
Dashboards, Logs
and Alerts



Splunk Forwarders
connected to the
Indexer



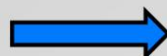
splunk.home.com
Ubuntu Server
24,10
192.168.1.109
Splunk SIEM
Logging and Alerting



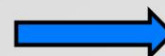
Dashboards, Logs
and Alerts



Wazuh Agents
Connected to
Indexer

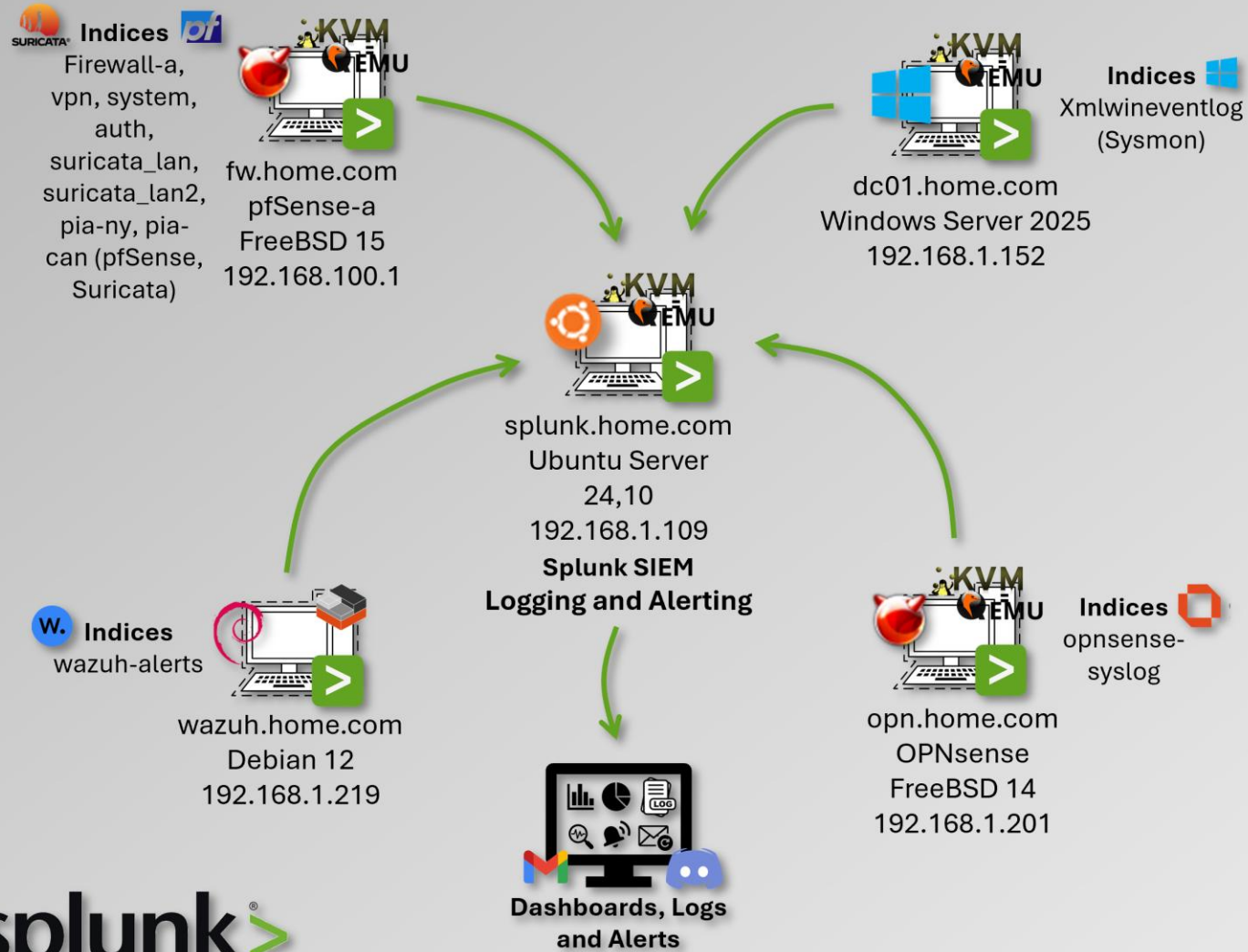


wazuh.home.com
Debian 12
192.168.1.219
EDR/XDR
FIM, SCA, Active Response,
Logging and Alerting

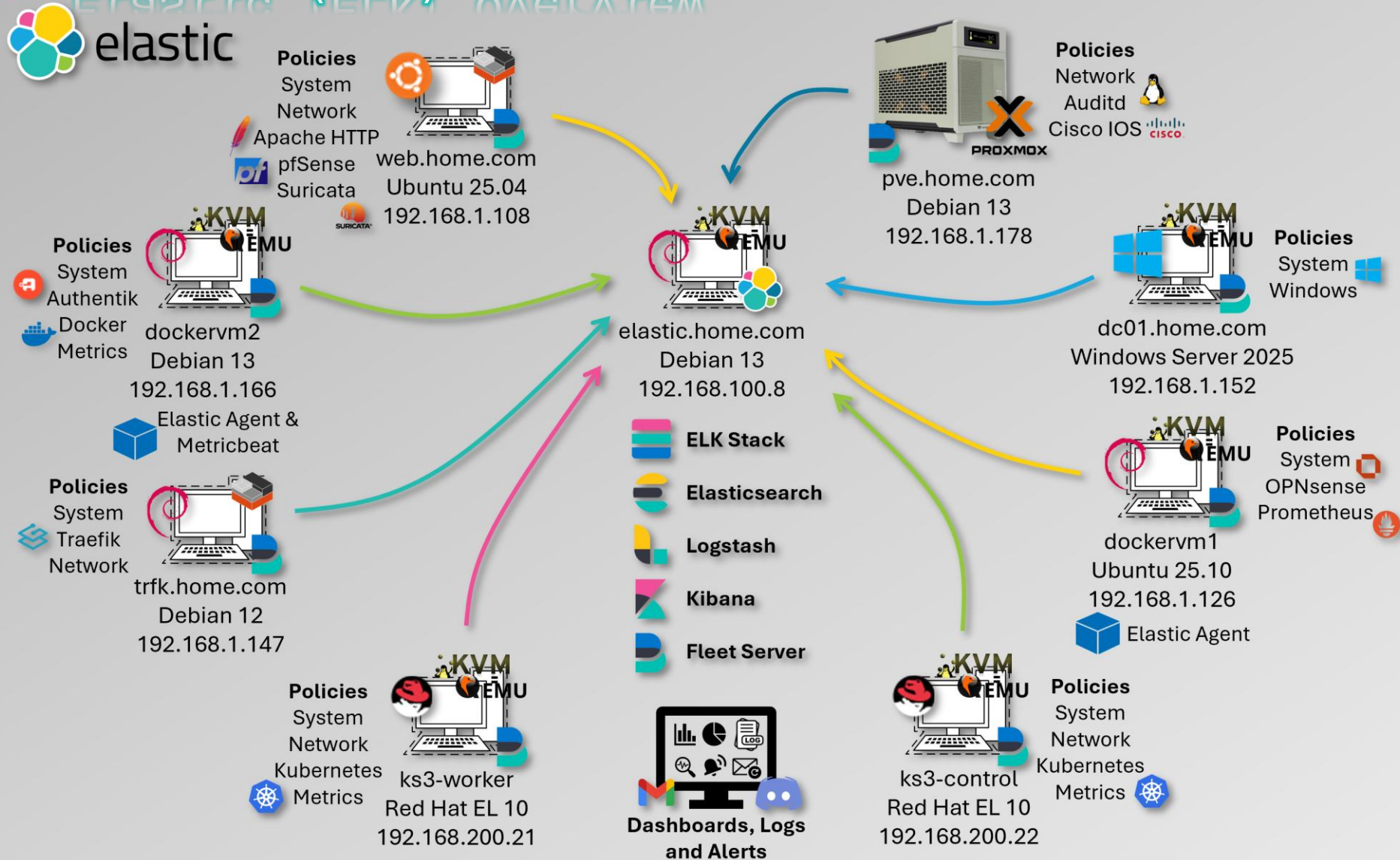


Dashboards, Logs
and Alerts

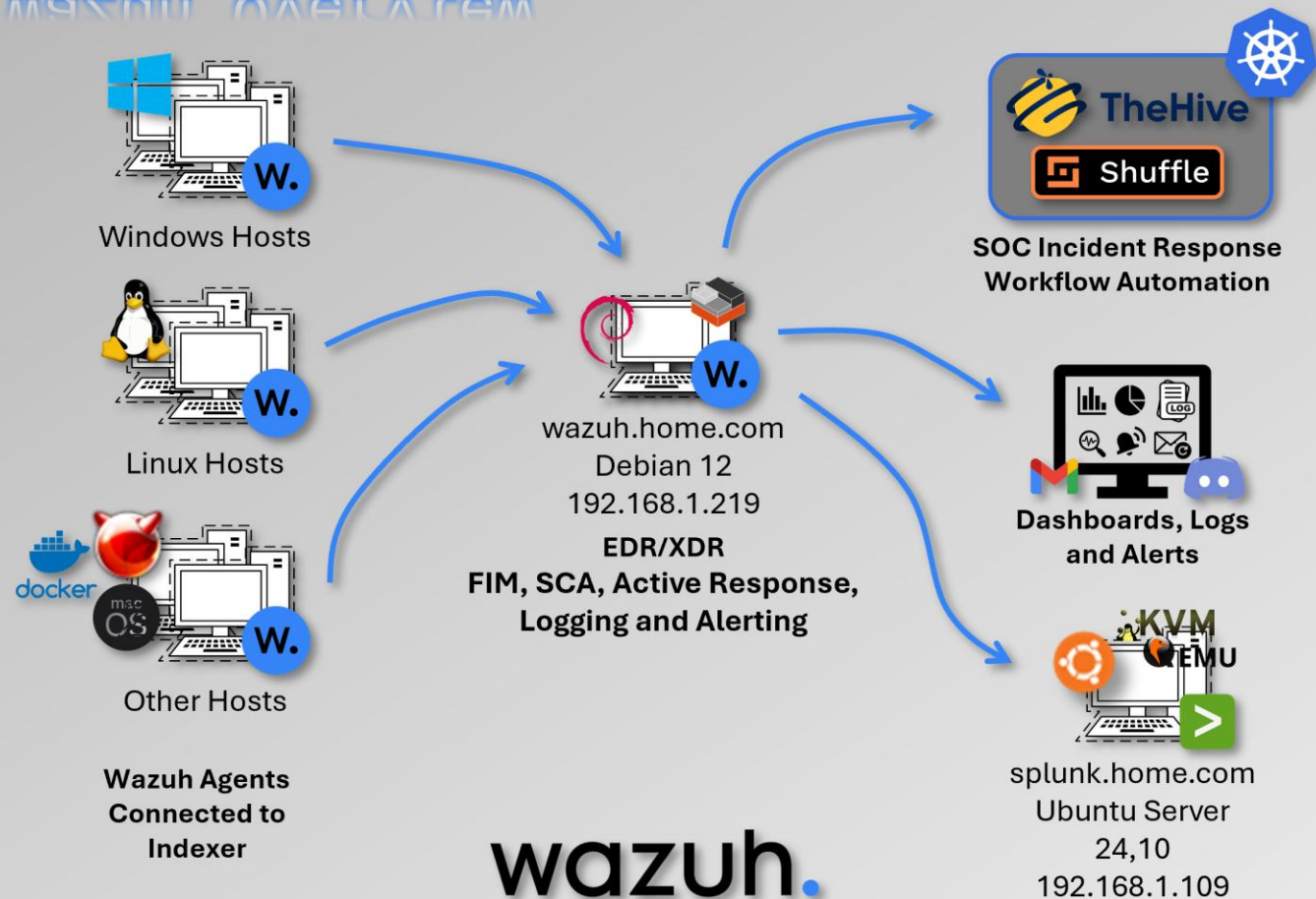
Splunk Overview



Elastic (ELK) Overview



Wazuh Overview



Shuffle SOAR Overview

Automated Incident Response Workflow wazuh.



Endpoint Alerts



wazuh.home.com
Debian 12
192.168.1.219



elastic.home.com
Debian 13
192.168.100.8



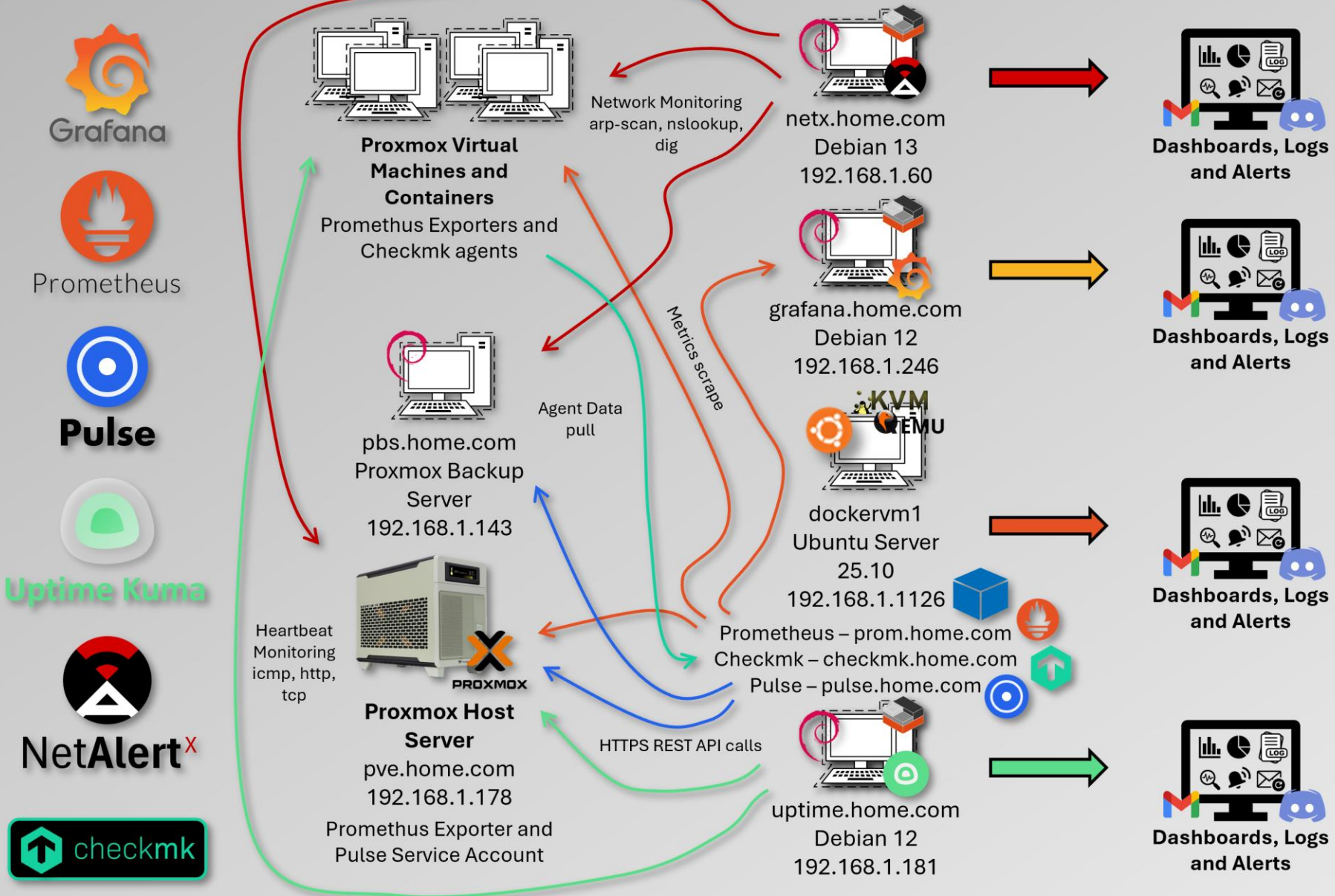
Network Alerts



Cases and Alerts



Monitoring Overview



Prometheus/Grafana Overview



Proxmox Virtual Machines and Containers

Prometheus Exporters for
Pihole, Node, Traefik,
Uptime Kuma, pfSense,
OPNsense and CrowdSec



Proxmox Host Server

pve.home.com
192.168.1.178
Prometheus Exporter



grafana.home.com
Debian 12
192.168.1.246



dockervm1
Ubuntu Server
25.10
192.168.1.126

Prometheus – prom.home.com
Alert Manager
Blackbox



Prometheus



Dashboards and
Alerts

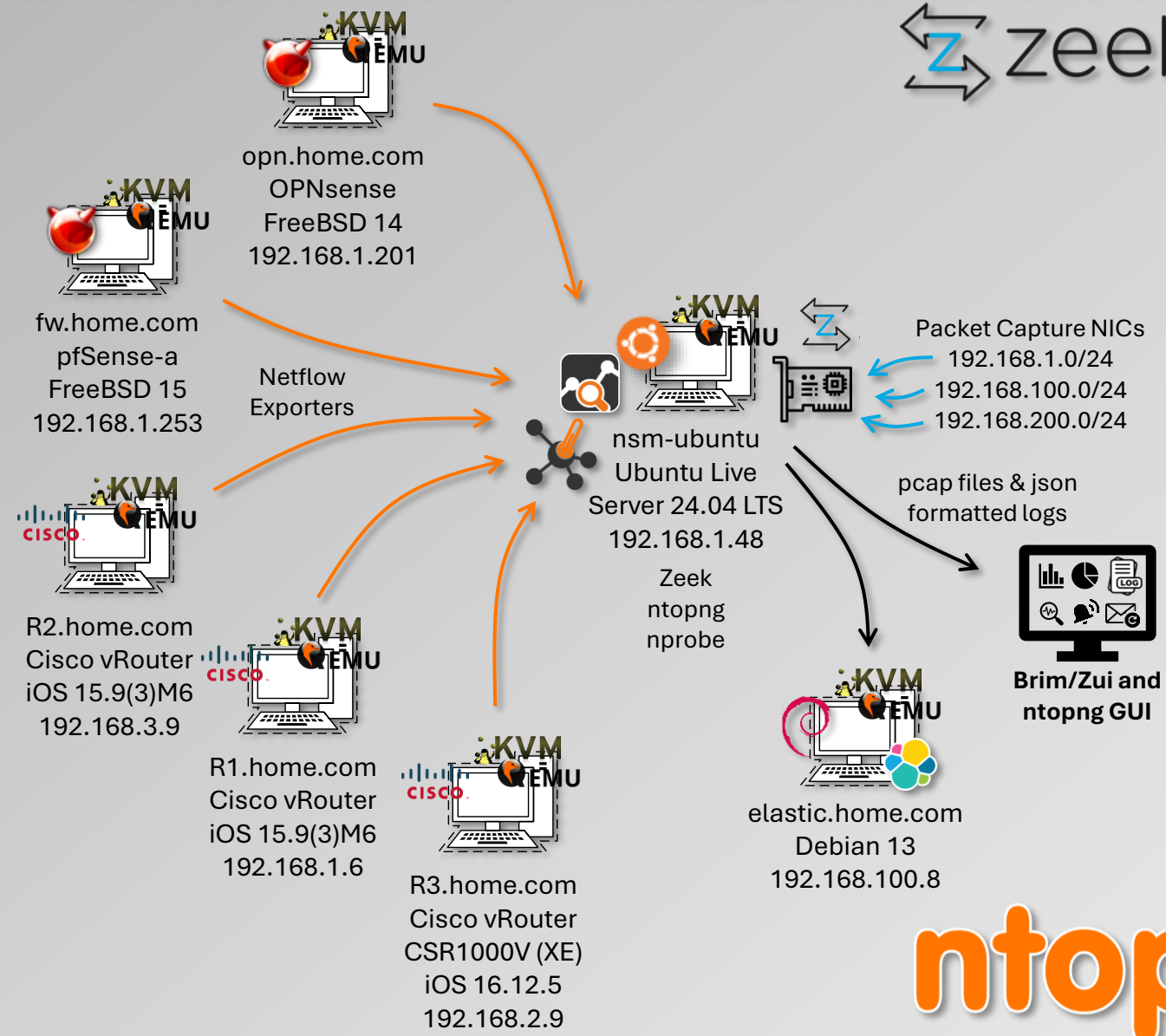


Grafana

Metrics
Scrapes

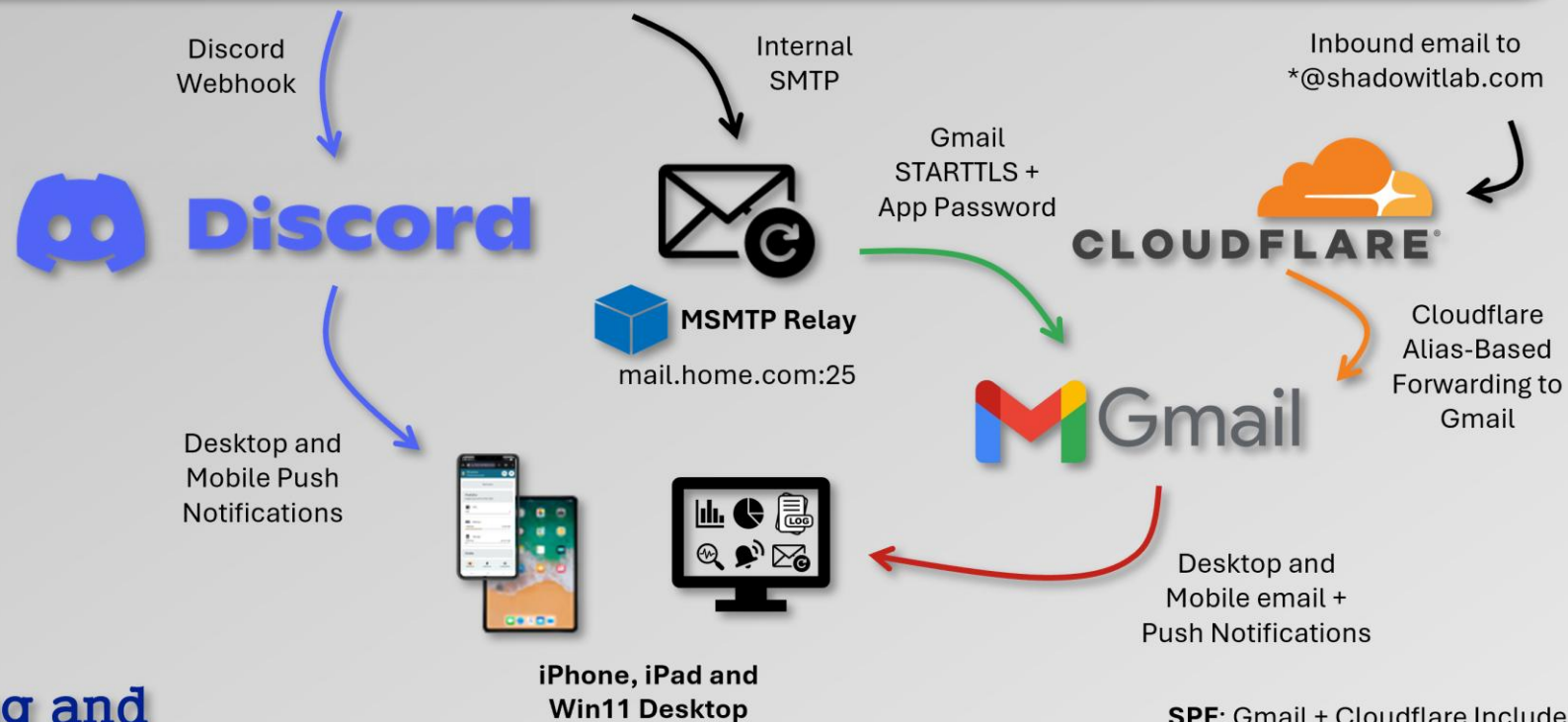
Alert
Notifications

Security Network Monitoring Overview



ntop

Notification Overview



Alerting and Notifications

SPF: Gmail + Cloudflare Include
DKIM: Gmail-signed Outbound, Cloudflare-signed Inbound
DMARC: Quarantine / Reject Policy